Vision-Language-Guided Waypoint Generation for Mapless Robot Navigation

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Motivation

Mobile robots usually navigate on a prebuilt map built from an occupancy grid and LiDAR points. In warehouses and other constantly changing environments, that map is not reliable. Objects move, so the global route is still planned on an outdated picture of the space. Each site also has to be mapped and annotated, and mapped again after every layout change. Even a perfect map is only geometry: it has no notion of a "second left" or an "elevator."

So we explore a different approach: the robot follows a spoken instruction, such as "go straight, take the second left, stop by the elevator," using what it observes at that moment, with no prebuilt map.

Our Approach

1. **Input.** A spoken multi-step instruction, plus the robot's live camera and LiDAR view. No prebuilt map.
2. **Output.** A short sequence of waypoints, not raw velocity commands.
3. **Execution.** Nav2 with MPC tracks the waypoints and avoids obstacles.

Language to Waypoints

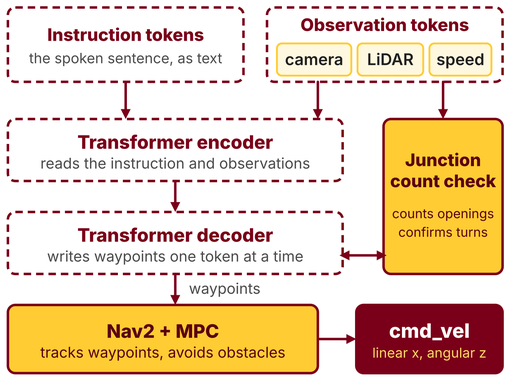


Fig. 1 The decoder emits waypoints from instruction, camera and LiDAR tokens, and Nav2 with MPC tracks them. Dashed: proposed. Gold: built or in progress.

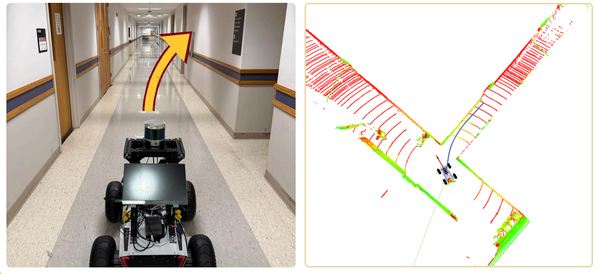
Experimental Test

"Take a left at the end of the hallway, then an immediate right. At the end of the path you will see an elevator, and near it a big red object. Stop next to it."

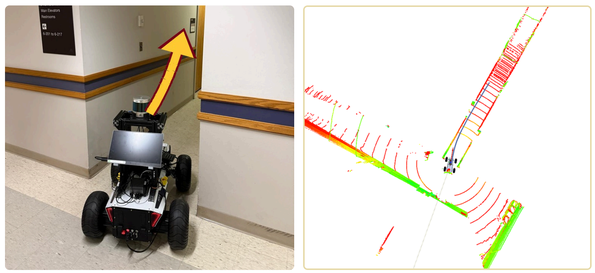
1 Hallway end, turn left



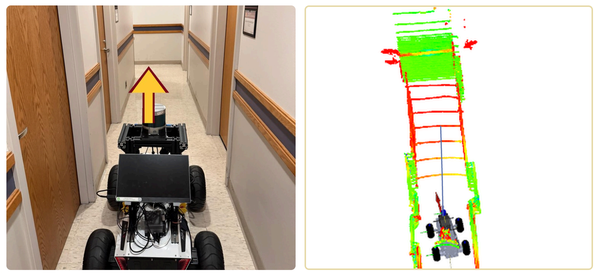
2 Immediate right



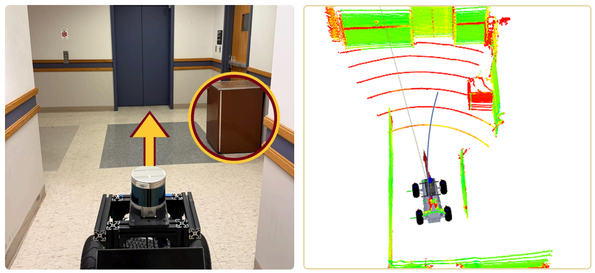
3 Into the passage



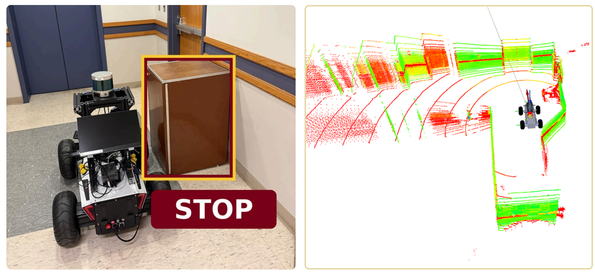
4 Follow it to the end



5 Elevator, red object



6 Stop next to it



Left: photo taken with the Hunter 2.0 in the test hallway, with the arrow marking the intended motion. Right: top-down RViz view of the 3D LiDAR point cloud at the same step.

Waypoint-Based Motion Planning

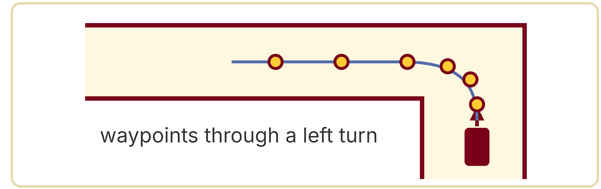


Fig. 2 Nav2 with MPC turns the waypoints into velocity commands (linear x, angular z) and steers around obstacles on the way.

Future Work

1. Train the waypoint decoder.
2. Ground any landmark named in the instruction.
3. Measure end-to-end success.
4. Test outdoors and in new layouts.

References

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[2] D. Shah, B. Osiński, B. Ichter, S. Levine, "LM-Nav: Robotic Navigation with Large Pre-Trained Models of Language, Vision, and Action," CoRL, 2022.

[3] C. Huang, O. Mees, A. Zeng, W. Burgard, "Visual Language Maps for Robot Navigation," ICRA, 2023.

[4] C. Simons, Z. Liu, B. Marcus, A. K. Roy-Chowdhury, K. Karydis, "Language-guided Robust Navigation for Mobile Robots in Dynamically-changing Environments," arXiv:2409.19459, 2024.

Acknowledgements

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